

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 23: Trigonometrical Ratios of Standard Angles

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Evaluate $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$.
2. Evaluate $\tan^2 60^\circ + 4 \cos^2 45^\circ + 3 \sec^2 30^\circ$.
3. Evaluate $\frac{\sin 30^\circ + \tan 45^\circ - \operatorname{cosec} 60^\circ}{\sec 30^\circ + \cos 60^\circ + \cot 45^\circ}$.
4. Evaluate $\cos^2 30^\circ - \sin^2 30^\circ$ and name the standard ratio it equals.
5. If $A = 30^\circ$, verify that $\cos 2A = 1 - 2 \sin^2 A$.
6. Find the acute angle θ if $\sqrt{3} \tan \theta = 1$.
7. Find θ if $\sin 2\theta = \frac{\sqrt{3}}{2}$ and 2θ is acute.
8. Find A and B if $\tan(A + B) = \sqrt{3}$ and $\tan(A - B) = \frac{1}{\sqrt{3}}$, with $A + B$ acute and $A > B$.
9. Solve $4 \cos^2 \theta = 3$ for an acute angle.
10. In triangle PQR, $\angle Q = 90^\circ$, $\angle P = 60^\circ$ and $PQ = 5$ cm. Find QR and PR.

Answer Key

1. 1

2. $3 + 2 + 4 = 9$

3. $\frac{\frac{3}{2} - \frac{2}{\sqrt{3}}}{\frac{3}{2} + \frac{2}{\sqrt{3}}} = \frac{43 - 24\sqrt{3}}{11}$

4. $\frac{1}{2} = \cos 60^\circ$

5. Both sides equal $\frac{1}{2}$

6. 30°

7. 30°

8. $A = 45^\circ, B = 15^\circ$

9. 30°

10. $5\sqrt{3}$ cm and 10 cm